TECHNICAL GUIDE

Box3D Metal Architecture

A compact guide to the Apple Silicon command graph, shared-memory ownership, constraint colors, and deterministic overflow.

APPLE SILICON / METAL COMPUTE / BOX3D OVERLAY

One ordered command graph

Supported constrained worlds encode every substep into one Metal command buffer, wait once, and read body state back once. This minimizes ownership changes on Apple unified memory.

Phase	Stage	Work
1	Integrate velocity	Forces, damping, gyroscopic term, speed limits
2	Warm start	Ordered overflow, then conflict-free graph colors
3	Solve	Biased contact and supported-joint iterations
4	Integrate position	Position and normalized quaternion update
5	Relax	Unbiased constraint iterations
6	Restitution	Eligible convex and mesh contacts
7	Finalize	Optional body state and awake-shape AABBs

Unconstrained worlds use a smaller fused path. Unsupported constrained worlds keep the CPU solver and may use only the independent Metal position stage.

Unified memory is not free synchronization

Persistent shared buffers avoid PCIe copies, but command submission, cache coherence, residency, and CPU/GPU ownership transitions still determine crossover.

Data layout and ownership

- Body state and compact body properties use geometrically grown persistent buffers.
- CPU contact preparation writes directly into shared Metal constraint allocations.
- Distance and parallel joints use separate compact type-dense buffers.
- Only accumulated joint solver state is unpacked after the command buffer.
- C/Metal size and offset assertions guard every shared ABI.
- Scalar shader vectors avoid native float3 padding surprises.

Graph colors

Constraints inside one non-overflow color never write the same dynamic body, so one GPU thread can own one constraint without atomics. Colors remain ordered with buffer barriers.

Deterministic overflow

Overflow constraints may share bodies. A single Metal thread walks them in upstream order. Mixed distance/parallel overflow uses an eight-byte type/index descriptor, preserving order with one launch per phase.

Experimental pair traversal

A separate Metal path traverses copies of Box3D's dynamic trees. A deterministic hierarchical SIMD scan assigns stable per-move offsets, then candidates are compacted in exact upstream tree and DFS order in one steady-state command buffer. CPU filters and contact creation remain unchanged. Bounded stack or capacity failures rerun the complete CPU traversal.

Apple GPU implementation choices

- One command buffer per solver step amortizes submission and synchronization.
- Persistent shared storage avoids redundant contact copies; pipeline-derived group widths follow device occupancy.
- Conflict-free colors avoid atomics, while body-sharing overflow stays serial and ordered.
- Safe Metal math reduces avoidable drift from the CPU oracle.